UNIT- 3

Dynamic Programming

The dynamic programming technique decomposes the original problem in n variables into n sub problems or n stages each in one variable.

The solution is obtained in an orderly manner by starting from one stage to the next and is completed after the final stage is reached.

Need of DP

1. It is required that all the decisions of a combination are specified.
2. The optimal policy can be selected only after all the combinations are evaluated.

(*) Bellman's principle of optimality?

An optimal policy has the property and that whatever be the initial state and initial decisions, the remaining decisions must constitute an optimal policy for the state resulting from the 1st decision.

(*) Characteristic of DPP?

1. The problem can be divided into stages, with the policy decision required at each stage.
2. The effect of the policy decision at each stage is to transform the current state into a state associated with the next stage.

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Applications of DP

1. In the production area, this technique has been used for Production, scheduling and employment smoothening problems.
2. It has been used to determine the interlory level and for formulating the inventory recording.
3. It is used to determine the optimal combination of advertising media.

1) Max $z = y_1 \cdot y_2 \cdots y_n$

$y_1 + y_2 + \ldots + y_n = c$

$y_i \geq 0 \quad i = 1, 2 \ldots n$

To develop the recursive equation

Let, y_i be the ith part of c and each i may be recorded as a stage.

The optimal solution at each stage are obtain by using usual classical method.

Let $f_n(c)$ be the maximum obtainable product

$y_1 \cdot y_2 \cdots y_n = c$

c is divided into n parts $y_1 \cdot y_2 \cdots y_n$

then $f_n(c)$ becomes a function of n

$f_n(c) = y_1 \cdot y_2 \cdots y_n$

for $n=1$ (one stage problem)

$f_1(c) = c$, trivial.

Here c is divided into one part only the is $y_1 = c$, Therefore $f_1(c) = c$

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$\max_{0 \le x \le c} f_1(c) = f.$

for n=2 (2 stage) : Here c is divided ind two parts

let $y_1 = x$, $y_2 = c - x$, then

$y_1 + y_2 = x + c - x = c$

$f_2(c) = \max\{y_1 \cdot y_2\}$

$0 \le x \le c$

$= \max\{x \cdot (c-x)\}$

$f_2(c) = \max\{x \cdot f_1(c-x)\}$

for n=3, (3 stage problem) Here c is divided into 3 parts y_1, y_2, y_3

let $y_1 = x_1$, $y_2 + y_3 = c - x$

$y_1 + y_2 + y_3 = c$

$(c-x)$ is further divided into two parts whose maximum attainable product $y_2 \cdot y_3$ is $f_2(c-x)$ then,

$f_3(c) = \max\{y_1 \cdot y_2 \cdot y_3\}$

$= \max\{x f_2(c-x)\}$

$f_n(c) =$

In general the recurrence equation for the n stage problem is

$f_n(c) = \max\{x f_{n-1}(c-x)\}$

$0 \le x \le c$

To solve the recursive equation,

for n=2

$f_2(c) = \max\{x \cdot f_1(c-x)\}$

the function

$x.(c-x)$ will be

maximum if, $f_2(c) = \max\{x(c-x)\}$

$$\frac{d}{dx}[x(c-x)] = 0$$

$$x(-1) + 1(c-x) = 0$$

$$-2x + c = 0$$

$$c = 2x$$

$$\boxed{x = c/2}$$

$$f_2(c) = \max\left\{\frac{c}{2}\left(c - \frac{c}{2}\right)\right\}$$

$$= \frac{c}{2}\left(\frac{c}{2}\right)$$

$$= \left(\frac{c}{2}\right)^2$$

$$op = \left\{\frac{c}{2}, \frac{c}{2}\right\}$$

optimal policy equal to $= \{c/2, c/2\}$

and $f_2(c) = (c/2)^2$

for $n=3$, we have

$$f_3(c) = \max\{x.f_2(c-x)\}$$

$$= \max\left\{x.\left(\frac{c-x}{2}\right)^2\right\}$$

The function $x.\left(\frac{c-x}{2}\right)^2$ will be maximum if

$$\frac{d}{dx}\left[x\left(\frac{c-x}{2}\right)^2\right] = 0$$

$$\frac{1}{4}\left[x.2(c-x)(-1) + 1(c-x)^2\right] = 0$$

$$(c-x)[-2x + c - x] = 0$$

$[c-x][c-3x] = 0$

$c-x=0$ (or) $c-3x=0$

$c = 3x$

$\boxed{x = c/3}$

$$f_3(c) = \max\left\{x\left(\frac{c-x}{2}\right)^2\right\}$$

$$= \max\left\{\frac{c}{3}\left(\frac{c-c/3}{2}\right)^2\right\}$$

$$= \frac{c}{3}\left(\frac{2c}{6}\right)^2$$

$$f_3(c) = (c/3)^3$$

optimal policy $= \{c/3, c/3, c/3\}$ & $f_3(c) = \left(\frac{c}{3}\right)^3$

Let's assume that the optimal policy for $n = m$ is

$$\left(\frac{c}{m}, \frac{c}{m} \cdots \frac{c}{m}\right) \text{ & } f_m(c) = \left(\frac{c}{m}\right)^m$$

now for $n = m+1$

$$f_{m+1}(c) = \max\{x\, f_m(c-x)\}$$

$$= \max\left\{x\left(\frac{c-x}{m}\right)^m\right\}$$

the function $x\left(\frac{c-x}{m}\right)^m$ attain its maximum

then $\frac{d}{dx}\left[x\left(\frac{c-x}{m}\right)^m\right] = 0$

we get $x = \frac{c}{m+1}$ and

$$f_{m+1}(c) = \left(\frac{c}{m+1}\right)^{m+1}$$

the result is true for $n = m+1$

Hence by mathematical induction, the optimal policy is $\left\{\frac{c}{n}, \frac{c}{n} \cdots \frac{c}{n}\right\}$ & $f_n(c) = \left(\frac{c}{n}\right)^n$

2. solve the following by DP

Min $z = y_1 + y_2 + \cdots + y_n$

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S.to c

$$y_1 \cdot y_2 \cdots y_n = b$$

$$y_1, y_2 \cdots y_n \geq 0$$

(or)

factorise a positive quantity b into n factors in such a way so that their sum is a minimum.

Sol:

To develop the recursive equation

Let $f_n(b)$ be the minimum attainable sum $y_1 + y_2 + \cdots y_n$

When the positive quantity b is factorised into n factors $y_1 \cdot y_2 \cdots y_n$

for $n=1$ (one stage problem)

Here b is factorised into one factor

$$y_1 = b$$

therefore $f_1(b) = b$ (trivial)

for $n=2$ (2 stage problem)

Here b is divided into two factors

$$y_1 = x, \quad y_2 = b/x$$

so that $y_1 \cdot y_2 = b$,

then, $\max = \{y_1 + y_2\}$

$$= \max_{0 \leq x \leq b} \left\{ x + \frac{b}{x} \right\}$$

$$= \max \left\{ x + f_1\left(\frac{b}{x}\right) \right\}$$

for $n=3$ (3 Stage prob)

b is factored into 3 factors

$y_1 = x \qquad y_2 \cdot y_3 = \frac{b}{x}$

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so that $y_1 \cdot y_2 \cdot y_3 = b$

$$f_3(b) = \max \{ y_1 + y_2 + y_3 \}$$

$$= \max \{ x + f_2\left(\frac{b}{x}\right) \}$$

In general the recursive equation for the n stage problem is

$$f_n(b) = \max \{ x + f_{n-1}\left(\frac{b}{x}\right) \}$$

solve recurrence equation.

for $n = 2$, we have

$$f_2(b) = \{ x + f_{n-1}\left(\frac{b}{x}\right) \}$$

$$f_2(b) = \max \{ x + b/x \}$$

the function $x + \frac{b}{x}$ will get maxim

if $f_2(b) = \max \{ x + \frac{b}{x} \}$

$$= \sqrt{b} + \frac{b}{\sqrt{b}}$$

$$= \frac{b + b}{\sqrt{b}} = \frac{2b}{\sqrt{b}}$$

$$f_2(b) = 2b^{1/2}$$

o.p $= \{ b^{1/2}, b^{1/2} \}, \quad f_2(b) = 2b^{1/2}$

$n = 3$ $\quad f_3(b) = \max \{ x + f_2(b/x) \}$

$$= \max \{ x + 2\left(\frac{b}{x}\right)^{1/2} \}$$

$$\frac{d}{dx}\left[x + \frac{2\sqrt{b}}{\sqrt{x}} \right] = 0$$

$1 + 2\sqrt{b}\left(-\frac{1}{2}\right)x^{-3/2} = 0$

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$$1 - \frac{\sqrt{b}}{x^{3/2}} = 0$$

$$x^{3/2} - \sqrt{b} = 0$$

$$x^{3/2} = b^{1/2}$$

$$x^3 = b$$

$$x = b^{1/3}$$

$$f_3(b) = b^{1/3} + 2\left(\frac{b}{b^{1/3}}\right)^{1/2}$$

$$= b^{1/3} + 2\left(\frac{b}{b^{1/3}}\right)^{1/2}$$

$$= b^{1/3} + 2b^{1/}$$

$$= b^{1/3} + 2b^{1/2} \cdot b^{(-1/3)(1/2)}$$

$$= b^{1/3} + 2b^{1/2} \; b^{-1/6}$$

$$= b^{1/3} + 2b^{4/2}$$

$$= b^{1/3} + 2b^{1/3}$$

$$f_3(b) = 3b^{1/3}$$

optimal policy $= \{b^{1/3}, b^{1/3}, b^{1/3}\}$

$f_3(b) = 3b^{1/3}$.

solve by DPP

$$\text{Min } z = y_1^2 + y_2^2 + \cdots + y_n^2$$

s.to.c $y_1, y_2, \cdots y_n = b$ &

$y_i \geq 0$ for all i

(or)

Factorise a positive quantity p into n factors in such a way that the sum of their squares is a minimum

soln: To develop the recursive equation

Let $f_n(b)$ be the minimum attainable sum

$$y_1^2 + y_2^2 + \cdots + y_n^2,$$

when the positive quantity b is factorised into n factors $y_1, y_2, y_3 \cdots y_n$

For $n=1$ (one stage problem)

$$\text{Min } z = y_1^2 + y_2^2 + \cdots + y_n^2$$

$f_n(b)$

$n=1$ $\quad y_1 = b$

$$f_1(b) = b^2$$

$n=2$

Let $y_1 = x$, $y_2 = b/x$

$$f_2(b) = \min_{0 \leq x \leq b} \{ y_1^2 + y_2^2 \}$$

$$= \min_{0 \leq x \leq b} \left\{ x^2 + \left(\frac{b}{x}\right)^2 \right\}$$

$$= \min \left\{ x^2 + f_1\left(\frac{b}{x}\right) \right\}$$

$n=3$

$y_1 = x, \quad y_2 \cdot y_3 = \frac{b}{x}$

$f_3(b) = \min\{y_1^2 + y_2^2 + y_3^2\}$

$= \min\{x^2 + f_2\left(\frac{b}{x}\right)\}$

$n=n$

$f_n(b) = \min\{x^2 + f_{n-1}\left(\frac{b}{x}\right)\}$

Solve for 2

$f_2(b) = \min\{x^2 + f_1\left(\frac{b}{x}\right)\}$

$= \min\{x^2 + \left(\frac{b}{x}\right)^2\}$

$\frac{d}{dx}\left\{x^2 + \left[\frac{b}{x}\right]^2\right\} = 0$

$2x - 2b^2 x^{-3} = 0$

$2x - \frac{2b^2}{x^3} = 0$

$x^4 - b^2 = 0$

$x^4 = b^2$

$x^2 = b$

$x = \pm\sqrt{b}$

$x = b^{1/2}$

opt. policy $= (b^{1/2}, b^{1/2})$

$f_2(b) = \min\left\{\left(b^{1/2}\right)^2 + \frac{b^2}{\left(b^{1/2}\right)^2}\right\}$

$= \frac{b^2 + b^2}{b}$

$= 2b$

$= 2\left(b^{1/2}\right)^2$

$f_3(b) = \min\left\{x^2 + f_2\left(\frac{b}{x}\right)^2\right\}$

$= \min\left\{x^2 + 2\left(\frac{b}{x}\right)\right\}$

$\frac{d}{dx}\left(x^2 + \frac{2b}{x}\right) = 0$

$2x - \frac{2b}{x^2} = 0$

$x = b^{1/3}$

op. policy : $b^{1/3}, b^{1/3}, b^{1/3}$

$f_3(b) = \left\{\left(b^{1/3}\right)^2 + \frac{2b}{b^{1/3}}\right.$

$= \frac{b^{2/3} \cdot b^{1/3} + 2b}{b^{1/3}}$

$= \frac{3b}{b^{1/3}}$

$= 3b^1 b^{-1/3} = 3b^{2/3}$

$= 3\left(b^{1/3}\right)^2$

optimal policy,

$\left(b^{1/n} \ldots b^{1/n}\right)$

$S_n(b) = n\left(b^{1/n}\right)^2$

4) Use DPP to solve the / show that (or)

$P_1 \log P_1 + P_2 \log P_2 + \ldots + P_n \log P_n$

$P_1 + P_2 + \ldots + P_n = 1 \qquad P_i \geq 0, \quad i = 1, 2, \ldots n$

- Is minimum when $P_1 = P_2 = \ldots P_n = 1/n$

(or)

Divide unity into n parts so, as to minimize the quantity $\sum P_i \log P_i$

soln: To develop the recursive equation

Let $f_n(1)$ be the minimum available sum $P_1 \log P_1 + P_2 \log P_2 + \ldots + P_n \log P_n$, when the unity is divided into n parts

$$P_1 + P_2 + \ldots + P_n = 1$$

for $n = 1$ (one stage problem)

$P_1 = 1$

$f_1(1) = 1 \log 1$

$n = 2$ Let $P_1 = x, \quad P_2 = 1 - x$

So, that

$P_1 + P_2 = 1$

$$f_2(1) = \min_{0 \leq x \leq 1} \{ x \log x + (1-x) \log (1-x) \}$$

$$= \min \{ x \log x + f_1(1-x) \}$$

$n = 3$ Let $P_1 = x, \quad P_2 + P_3 = 1 - x$, so that

$P_1 + P_2 + P_3 = 1,$

$$f_3(1) = \min \{ x \log x + f_2(1-x) \}$$

for n

$$f_n(1) = \min\{x \log x + f_{n-1}(1-x)\}$$

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To solve R.E.

for $n=2$

$$f_2(1) = \min\{x \log x + f_1(1-x)\}$$

$$= \min\{x \log x + (1-x)\log(1-x)\}$$

To get n $\quad \frac{d}{dx}\left[x \log x + (1-x)\log(1-x)\right] = 0$

$$\frac{x}{x} + \log x + \frac{1-x}{1-x}(-1) + (-1)\log(1-x) = 0$$

$$1 + \log x - 1 - \log(1-x) = 0$$

$$\log \frac{x}{1-x} = 0$$

$$e^{\log \frac{x}{1-x}} = e^0$$

$$\frac{x}{1-x} = 1$$

$$x = 1-x$$

$$x + x = 1$$

$$2x = 1$$

$$x = 1/2$$

op.pt $\{1/2, 1/2\}$

$$\boxed{f_2(1) = 1/2}$$

op.poly. $\{\frac{1}{3}, \frac{1}{3}, \frac{1}{3}\}$

$$f_n(3) = 3\left(\frac{1}{3}\log\frac{1}{3}\right).$$

$$f_2(1) = \frac{1}{2}\log\frac{1}{2} + \left(1-\frac{1}{2}\right)\log\left(1-\frac{1}{2}\right)$$

$$= 2\left(\frac{1}{2}\log\frac{1}{2}\right)$$

1) To develop the recursive equation.

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It is a 3 stage problem.

The decision variables are x_1, x_2, x_3 and state variables are S_1, S_2 and S_3 are defined as

$$\text{Min } z = x_1^2 + x_2^2 + x_3^2$$

S to, $x_1 + x_2 + x_3 \geq 15$ & $x_1, x_2, x_3 \geq 0$

$$S_3 = x_1 + x_2 + x_3 \geq 15$$

$$S_2 = x_1 + x_2 = S_3 - x_3$$

$$S_1 = x_1 = S_2 - x_2$$

$$f_1(S_1) = \min z\{x_1^2\}$$
$$= S_1 = (S_2 - x_2)^2$$

$$f_2(S_2) = \min\{x_1^2 + x_2^2\}$$
$$= \min\{x_2^2 + f_1(S_1)\}$$
$$= \min\{x_2^2 + (S_2 - x_2)^2\}$$

$$f_3(S_3) = \min\{x_1^2 + x_2^2 + x_3^2\}$$
$$= \min\{x_3^2 + f_2(S_2)\}$$

Solve R.E

$$f_1(S_1) = S_1 = (S_2 - x_2)^2$$

$$f_2(S_2) = \min\{x^2 + f(S_1)\}$$
$$= \min\{x_2^2 + (S_2 - x_2)^2\}$$
$$= \frac{d}{dx_2}\{x_2^2 + (S_2 - x_2)^2\} = 0$$
$$\frac{d}{dx_2}\{x_2^2 + S_2^2 + x_2^2 - 2S_2 x_2\} = 0$$

$2x_2 + 2x_2 - 5S_2 = 0$

$4x_2 = 2S_2$

$$\boxed{x_2 = \frac{S_2}{2}}$$

$$f_2(S_2) = \min\left\{\frac{S_2^2}{4} + \left(S_2 - \frac{S_2}{2}\right)^2\right\}$$

$$= \min\left\{\frac{S_2^2}{4} + \frac{S_2^2}{4}\right\}$$

$$= \frac{2S_2^2}{4} = \frac{S_2^2}{2}$$

$$f_3(S_3) = \min\left\{x_3^2 + \frac{S_2^2}{2}\right\}$$

$$= \min\left\{x_3^2 + \frac{(S_3 - x_3)^2}{2}\right\}$$

$$\frac{d}{dx_3}\left[x_3^2 + \frac{(S_3 - x_3)^2}{2}\right\} = 0$$

$$2x_3 + \frac{2(S_3 - x_3)(-1)}{2} = 0$$

$$2x_3 - \frac{2S_3}{2} + \frac{2x_3}{2} = 0$$

$$3x_3 - S_3 = 0$$

$$3x_3 = S_3$$

$$\boxed{x_3 = \frac{S_3}{3}}$$

$$f_3(S_3) = \min\left\{\frac{S_3^2}{9} + \frac{\left(S_3 - \frac{S_3}{3}\right)^2}{2}\right\}$$

$$= \min\left\{\frac{S_3^2}{9} + \frac{\left(\frac{2S_3}{3}\right)^2}{2}\right\}$$

$$= \min\left\{\frac{S_3^2}{9} + \frac{4S_3^2}{18}\right\}$$

$= \min \left\{ \frac{S_3^2}{9} + \frac{4 S_3^2}{18} \right\}$

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$= \min \left\{ \frac{6 S_3^2}{18} \right\} = \min \frac{S_3}{3}$

$S_3 = 15,$

$x_3 = 15/3 = 5$

$S_2 = S_3 - x_3 \Rightarrow 15 - 5 = 10$

$x_2 = \frac{S_2}{2} = \frac{10}{2} = 5$

$S_1 = x_1 = S_2 - x_2$

$= 10 - 5 = 5$

optimal $\{x_1, x_2, x_3\}$.

The optimal policy is $\{x_1, x_2, x_3\} = \{5, 5, 5\}$

Min z function $= 25 + 25 + 25 = 75$

$x_1^2 + x_2^2 + x_3^2$

2) Use DP solve

max $= y_1 \cdot y_2 \cdot y_3$

s.to: $y_1 + y_2 + y_3 = 5$ and $y_1, y_2, y_3 >= 0$

soln: Develop recursive equation,

The decision variables are y_1, y_2, y_3 and the state variables s_1, s_2, s_3 are defined as,

max $= y_1 \cdot y_2 \cdot y_3$

$s_3 = y_1 + y_2 + y_3 = 5$

$s_2 = y_1 + y_2 = s_3 - y_3$

$s_1 = y_1 = s_2 - y_2$